Outline and justification of software engineering methods (3 marks, 2 pages)

We have used google drive as our collaboration tool for documentation. This is because we are all familiar with it, and it is suitable for version control due to the ability to view document history. We have also decided to use github for the implementation aspect of the project, being able to keep all of the parts of the code together. We have used libgdx as the game engine for the project, as it is the one with the most support available and the one that we deemed would be the easiest to get to grips with as a team. We have decided we will use generative AI to help with the graphical assets, as this will help to save time with things such as the map of the campus, as this needs to be abstracted to create something usable for a game of this level. It will also help with things such as characters and items, as it will help us to keep a consistent style throughout the game. This is also useful as we have been told we need to be wary of licenses when creating this game, so this means we can avoid the risk of using images that bring licensing issues.

After the meeting with the customer, we have a better grasp on what is required out of this game. It is a game targeted for undergraduate students, and the game will be used with prospective students for open days in order to demonstrate to them what they will be able to achieve during their studies at the university. The game should be compatible with desktop computers, as well as laptops. This means the screen size for the game shouldn't be hard coded, and it should be able to run on devices with screens ranging from 13 inch to 32 inch. We also discovered that when making the game, we need to take accessibility into consideration, and colourblindness is a big factor that should be taken into consideration when making the game. Some research needs to be conducted on our end in order to properly cater for this particular accessibility requirement. It has been made very clear that it is in our best interest to avoid complicated features, as this is unnecessary, and with such a short timescale for the development of the game, it is important to streamline and prioritise the important features of the game. Things such as music (and therefore volume controls), colour schemes, keybinds/controls, menus etc. should be not only intuitive to the user, but also should be clearly labeled in order to avoid confusion.

Outline of the teams approach to organisation (2 marks, 1 page)

We have chosen to communicate via whatsapp, as this allows us to message about work, as well as hold meetings through whatsapp calls when people are unable to meet in person. Regarding meetings, we have planned to have meetings as needed, taking advantage of the weekly labs to ensure we have at least 1 meeting a week. As the project goes on, we will need to have more meetings to ensure we stay on top of work. We are using a shared google drive, containing shared google docs in order to complete the documentation of the work, this way we can see how each member is getting on with their assigned work, as well as view the edit history of the documents to see when each member has made an amendment to the document. It also allows us to view previous versions of the documents should we want to revert to a previous point in the project. We have a shared github project where all the coding is contained, allowing us all to view it and contribute where necessary. Regarding workload allocation, we have assigned each member to a role in order to effectively complete the project. This is as follows:
Website (submit only the URL) **3** — url1.txt
1 - laith

Requirements **20** 1 + 3 Req1.pdf

1 - Elliot

Architecture **22** 6 Arch1.pdf

2 - Arthur, Somil

Method selection and planning **10** 2 + 1 + 2 Plan1.pdf

1 - Noah

Risk assessment and mitigation **10** 1 + 3 Risk1.pdf

1 - laith

Implementation **25**

2 - Marc and Marcus

Each member has been assigned roughly 12 marks worth of work. Each member has been assigned to an area which they feel they can contribute effectively to. Should members be unable to contribute for a set period of time due to unforeseen events such as illness, we will look to reassign this work.

Systematic plan for the project (5 marks, 2 pages)

Week 1: we came up with questions to ask the customer during our meeting with them in order to best inform our project. We also came up with a few different ideas for the game, such as different unexpected events and different features. Start date: 6/10/2026 end date: 14/10/2026 (date of the meeting with the customer)
Week 2: we assigned each member of the group to a part of the project, ensuring each member is responsible for approximately 12 marks. At this point we still needed to have the meeting with the customer, so we weren't able to proceed much further with the project.
Week 3: at this point the meeting with the customer had happened, and we had a better idea of what was required from us during this project. We reassigned people to different areas of the project due to preference.